

# PROTOKOL

Quality Traceability for Public Infrastructure  
Technical Product Document — v2.0

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Document   | Technical Product Document (what to build, why, and what we found) |
| Version    | v2.0 — supersedes v1.0                                             |
| Date       | September 2026                                                     |
| Written by | Kenny Garamendi — Product                                          |
| For        | Snehil Sahay — Engineering                                         |
| Market     | Public road construction, Peru                                     |
| Status     | Pre-build. Nothing has been coded yet.                             |

*Read Section 2 first. Without it, the rest of this document is just words.*

## 1. EXECUTIVE SUMMARY

We have privileged access to a live public infrastructure project in Peru — a S/ 41.9M (≈ USD 11M) road construction contract — including its full document archive, its internal coordination channel, and its quality manager. We spent six weeks analyzing 487 files and the complete team chat history.

What we found is a documentation system that is legally mandatory, financially blocking, and completely broken. Quality records that the law requires are being kept in paper notebooks, WhatsApp messages, and Excel files that contradict each other. Without those records, the contractor cannot get paid, and the quality engineer personally carries legal liability for seven years after delivery.

Protokol is the system that fixes this. It captures quality data at the point where work happens, validates it against the legally binding technical standard, and produces the document package the state requires — automatically.

### Why this is worth building

| Factor                             | Evidence                                                                                                                                                      |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The problem is real and documented | We have the quality manager writing in all caps in the team chat: "AS OF TODAY WE ARE NOT MANAGING TO COMPLETE THE PROTOCOLS"                                 |
| The market is large                | 2,741 public works projects are currently stalled in Peru, representing over S/ 67 billion in committed investment. Documentation failure is a leading cause. |
| We have insider access             | Full access to a live project's document archive, team chat, and processes — most competitors build blind                                                     |
| There is a paying trigger          | Missing quality documentation blocks the monthly payment certificate. This is not a nice-to-have.                                                             |
| We have a commercial channel       | A civil engineer with sector reputation and institutional contacts is ready to sell this using his name and network                                           |

### Who is involved

| Person             | Role                                        | Relationship                        |
|--------------------|---------------------------------------------|-------------------------------------|
| Kenny Garamendi    | Product, research, go-to-market             | Founder                             |
| Snehil Sahay       | Engineering, architecture, implementation   | Founder — full technical authority  |
| David Valdez Ochoa | Domain expert and commercial channel. Civil | Not a founder. Commercial partner — |

|  |                                                                                                           |                                 |
|--|-----------------------------------------------------------------------------------------------------------|---------------------------------|
|  | engineer, currently Quality Specialist on the pilot project. Has sector contacts and institutional reach. | terms to be defined separately. |
|--|-----------------------------------------------------------------------------------------------------------|---------------------------------|

## 2. HOW PERUVIAN PUBLIC CONSTRUCTION ACTUALLY WORKS

► *Snehl: this section exists because you cannot design this system correctly without understanding the physical and legal process it models. Read it slowly. Everything else depends on it.*

### 2.1 The physical process — how a road gets built

A concrete road is not poured in one go. It is built in horizontal layers, and each layer must be approved before the next one can go on top. This sequencing is the core of the system we are building.

| Step             | What happens physically                                               | What must be verified                                                     |
|------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1. Subgrade      | The natural soil is cut, leveled and compacted to form the foundation | Compaction density, moisture content, elevation                           |
| 2. Sub-base      | A granular material layer (20cm) is spread and compacted              | Compaction $\geq 100\%$ of lab reference, thickness, material certificate |
| 3. Base          | A second granular layer (25cm), higher quality                        | Same as sub-base, plus load-bearing test                                  |
| 4. Formwork      | Wooden or metal molds are set to contain the concrete                 | Dimensions, alignment, elevation                                          |
| 5. Steel         | Reinforcement bars are placed inside the formwork                     | Bar diameter, spacing, concrete cover                                     |
| 6. Concrete pour | Ready-mix concrete arrives by truck and is poured into the formwork   | Consistency test on arrival, sample cylinders taken, truck ID logged      |
| 7. Curing        | The fresh concrete is kept wet for days so it develops strength       | Curing method and duration                                                |
| 8. Strength test | The sample cylinders are crushed in a lab at 7 and 28 days            | Compressive strength must meet design value                               |

**Key insight for architecture: this is a state machine over physical segments, not a CRUD app over forms.**

### 2.2 The players — and who actually wants this product

| Actor                              | What they do                                                             | Incentive regarding Protokol                                                 |
|------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Field technician                   | Executes and measures on site. Sweat, dust, phone in pocket.             | Neutral to negative. More work unless it is faster than what they do now.    |
| Quality Specialist                 | Signs off that everything complies. Personally liable for 7 years.       | STRONGLY POSITIVE. This is our user and our champion. David holds this role. |
| Site Resident Engineer             | Runs execution. Needs the monthly payment certificate approved.          | Positive. Missing documents delay his payment.                               |
| Supervisor / Inspector             | Independent auditor appointed by the state entity. Verifies and objects. | Positive. Less time chasing paperwork.                                       |
| State entity (Regional Government) | Owens the project and the budget. Audited by the Comptroller.            | Positive at institutional level, slow at decision level.                     |
| Comptroller General                | National audit body. Can sanction individuals.                           | Not a customer, but the reason the customer needs us.                        |

### 2.3 The legal and financial machinery

Three facts drive the entire business case. Internalize these:

## Fact 1 — Quality documents gate the money

Contractors get paid monthly through a document called a valorización (payment certificate). It measures physical progress and requests payment. If the quality documentation backing that progress is incomplete, the certificate is objected and payment is delayed. We have the internal memos from this project proving this — the Site Resident formally requesting quality certificates "for evaluation, conformity and payment processing."

## Fact 2 — Personal liability lasts seven years

Under Peruvian Civil Code Article 1784, if a structure shows serious defects or partial collapse, the builder is liable. For public works, that liability extends a minimum of seven years from formal acceptance. The quality documentation is the only defense. A quality engineer with no records is personally exposed.

## Fact 3 — There is already a mandatory state system, and it is a wedge, not a competitor

Since June 2024, Comptroller Directive N° 017-2023-CG/GMPL governs projects executed directly by state entities. It requires the resident and supervisor to register execution information in INFOBRAS — the Comptroller's public works information system — replacing the old paper site logbook. Registration is mandatory under sanctionable responsibility.

- This is our strongest commercial angle and it must shape the architecture: Protokol should feed INFOBRAS, not compete with it. Design the data model so that exporting to INFOBRAS format is straightforward.

## 2.4 Glossary — terms used throughout this document

| Spanish term             | English meaning                                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------------------------------|
| Progresiva               | Chainage. Distance along the road centerline, written as km+meters. '0+144' = 144 meters from start.                 |
| Paño                     | Slab panel. A rectangular section of pavement poured in one work session. The daily unit of work.                    |
| Vaciado                  | Concrete pour.                                                                                                       |
| Encofrado                | Formwork. The temporary mold holding fresh concrete.                                                                 |
| Slump                    | Concrete consistency test. A cone is filled, lifted, and the mix subsidence is measured in cm. Checks workability.   |
| f <sub>c</sub>           | Design compressive strength of concrete, in kg/cm <sup>2</sup> . Peru uses kg/cm <sup>2</sup> , not MPa or psi.      |
| Probeta / Testigo        | Concrete test cylinder, cast during the pour, crushed at 7 and 28 days.                                              |
| Mixer                    | Ready-mix concrete truck.                                                                                            |
| Curado                   | Curing. Keeping concrete moist so it develops strength.                                                              |
| Subrasante               | Subgrade. Prepared natural soil layer.                                                                               |
| Sub base / Base granular | Granular sub-base and base layers.                                                                                   |
| Proctor Modificado       | Modified Proctor. Lab test establishing maximum achievable soil density. Field compaction is expressed as % of this. |
| Cota                     | Elevation level, verified by survey equipment.                                                                       |
| Protocolo                | Quality protocol. The formal record proving an activity met standard. Our core artifact.                             |
| Liberación               | Release / sign-off allowing work to proceed to the next stage.                                                       |
| No Conformidad (NC)      | Non-conformance. Formal record that something failed a criterion. Requires documented corrective action.             |
| Dossier de Calidad       | Quality dossier. The complete compiled documentation package. Required to close the contract.                        |
| Valorización             | Monthly payment certificate based on measured progress.                                                              |
| Expediente Técnico       | Master project document: drawings, specifications, quantities, budget.                                               |
| Metrado                  | Quantity take-off.                                                                                                   |
| Residente de Obra        | Site Resident Engineer — runs execution.                                                                             |
| Supervisor / Inspector   | Independent supervising engineer appointed by the state entity.                                                      |
| Administración Directa   | Direct execution. The state entity builds with its own staff rather than contracting a construction company.         |
| EG-2013                  | Ministry of Transport road construction specification manual. Legally binding technical standard.                    |
| INFOBRAS                 | Comptroller General's mandatory public works information system.                                                     |

### 3. WHAT WE INVESTIGATED AND HOW

This document is not based on assumptions about the market. It is based on primary evidence from a live project. Here is exactly what we analyzed:

| Source                               | Volume                                                      | What it gave us                                                                                               |
|--------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Project quality Drive (full archive) | 487 files across 49 folders, 6.79 MB of extracted text      | The official document taxonomy, all procedures, the master protocol tracker, technical specifications, budget |
| Team WhatsApp group                  | Complete history, 31 Jul – 27 Aug 2026                      | How work is actually coordinated, real frustrations, response patterns                                        |
| Official memoranda                   | 6 formal memos between Site Resident and Quality Specialist | Proof that quality documents gate payment; contract number and legal framing                                  |
| Monthly quality report               | Report N°01-2026, written by the Quality Specialist         | Complete project history: 10 time extensions, budget growth, and the state of quality management on arrival   |
| Master protocol tracker              | 84 rows of live quality records                             | Quantified data inconsistencies                                                                               |
| Technical file                       | Reformulated budget, drawings, quantity take-offs           | Real activities, materials and specifications                                                                 |
| Concrete mix designs                 | Lab reports from the accredited testing lab                 | The actual validation thresholds for this project                                                             |

#### The pilot project

| Attribute            | Value                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------|
| Project              | Road transitivity improvement and expansion, section AY-728 to AY-729                         |
| Location             | Ayacucho region, Peru                                                                         |
| Length               | 2.38 km                                                                                       |
| Contract             | N° 81-2026-GRA-SEDECENTRAL-OAPF                                                               |
| Owner                | Regional Government of Ayacucho                                                               |
| Execution mode       | Direct execution (Administración Directa)                                                     |
| Original budget      | S/ 16.3M — reformulated to S/ 41.9M (+156%)                                                   |
| Original duration    | 240 days — extended to 1,325 calendar days                                                    |
| Time extensions      | 10 (AP01 through AP10)                                                                        |
| Progress at analysis | 53.19% physical, 68.37% financial                                                             |
| Scope                | Rigid pavement, asphalt, a bridge, curbs, sidewalks, culverts, gravity walls, retaining walls |

### 4. FINDINGS — WHAT IS ACTUALLY BROKEN

Every finding below is backed by a specific artifact we hold. These are not hypotheses.

#### Finding 1 — Quality records are not being completed, and everyone knows it

On 8 August 2026, the Quality Specialist wrote to the entire team group, in capital letters:

**"AS OF TODAY WE ARE NOT MANAGING TO COMPLETE THE PROTOCOLS. I NEED YOU YOURSELVES TO BE PREPARING THE PROTOCOLS AND FILLING THEM IN."**

No one replied. The next messages in the chat are unrelated photos. Earlier the same day another team member wrote that there were no protocol forms left and they needed to be printed.

► *Interpretation that matters for design: the bottleneck is not awareness. Everyone knows the protocols are needed. The bottleneck is friction at the moment of recording.*

#### Finding 2 — There are five parallel sources of truth for the same data

The same construction reality is recorded in at least five disconnected places, none of which reconcile automatically:

- The pour schedule (Excel) — which panel gets poured on which day
- The ready-mix delivery log (Excel, 137 rows) — one row per concrete truck
- The lab test template (Excel) — sample cylinders and results
- A physical paper notebook — cement bags consumed per structural element
- The master protocol tracker (Excel) — quality release status

Concrete evidence of divergence: we hold two files with the same name, Cronograma\_Progresivas.xlsx, shared days apart in the same chat. For the same road section on the same day, one states 96.5 m<sup>3</sup> of concrete required and the other states 39.9 m<sup>3</sup>. One of them is wrong, and nobody knows which.

Additionally, the same technical drawing exists in four formats: the native CAD file, a clean PDF exported from it, a photocopier scan of a printed copy, and a re-shared duplicate under a different filename. We verified two of these files are byte-for-byte identical with different names.

### Finding 3 — Paper loses data permanently

The cement consumption notebook records bags used per structural element per date. On one entry, dated 24/07, the quantity field contains a handwritten question mark. The person recording did not know the number and it was never recovered. That data is gone forever.

### Finding 4 — The tracker itself contains contradictions

We analyzed the master protocol tracker (84 rows). Two systematic problems:

| Issue                                                        | Count            | Implication                                                                                                                            |
|--------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Rows showing progress above 100%, some reaching 154%         | 26 of 84 rows    | The baseline denominator is wrong or unmaintained. The percentages are meaningless.                                                    |
| Rows with recorded non-conformances still marked as COMPLIED | 17 rows affected | Either the sign-off criterion ignores the non-conformance column, or people are marking status without checking. Both are audit risks. |

### Finding 5 — Procedures were copied from an unrelated project and never adapted

We reviewed three work procedure documents from the project's own procedures folder. All three have a correct cover page naming this project — and a body that still refers to a completely different project: a mining project executed by different companies in 2011–2012. Internal document codes still carry the other project's prefix.

► *This is a serious document-control finding in a state contract. It is also a product opportunity: generated documents that are always correctly attributed are a feature worth paying for.*

### Finding 6 — The existing digital tool was bought and abandoned

The project has a construction management platform installed and paid for. The team does not use it. All real coordination — test requests, photos, schedules, mix designs, even the memoranda — flows through WhatsApp.

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***This is the single most important lesson in this document: a tool that requires people to change channel will be abandoned, regardless of how good it is.***

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### Finding 7 — The quality system did not exist until three months ago

According to the Quality Specialist's own first monthly report, when he joined the project in June 2026 — with the project already nearly four years into execution — he found: no quality plan, no construction procedures implemented, no quality dossier, and no non-conformance register. Zero. He also identified physical defects in already-built structures: cracking in two retaining walls and segregation voids in several concrete walls.

## 5. THE OPPORTUNITY

### 5.1 Why a business exists here

Three forces converge:

- Legal obligation — quality documentation is mandatory, not discretionary. Under the Comptroller directive, registration carries sanctionable responsibility.
- Financial trigger — incomplete documentation delays payment certificates. There is immediate cash pain.
- Personal exposure — the quality engineer is individually liable for seven years. The buyer has skin in the game.

Most construction software sells efficiency. Protokol sells protection and cash flow. That is a fundamentally stronger sale.

### 5.2 Market size

| Metric                                                  | Figure                                                                  | Source                            |
|---------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------|
| Stalled public works projects in Peru                   | 2,741 as of early 2026                                                  | Comptroller General               |
| Committed investment in stalled projects                | Over S/ 67 billion                                                      | Comptroller General               |
| Leading cause cited                                     | Insufficient technical capacity in project management and documentation | Comptroller General               |
| Audited projects that doubled cost and tripled schedule | 32 in a single 2026 audit                                               | Comptroller General               |
| Our pilot project's own record                          | +156% budget, 10 time extensions                                        | Project's internal monthly report |

The relevant addressable market is every public infrastructure project executed under direct administration in Peru. Ayacucho region alone runs dozens simultaneously. The same regulatory framework applies in all 25 regions.

### 5.3 Competitive position — honest assessment

International construction management platforms exist and are available in Peru. Digital checklist products with mobile photo capture and signatures also exist in the Spanish-speaking market. We are not the first to build digital site forms.

Our defensibility is not the software. It is three things that compound:

| Asset                       | Why it is hard to copy                                                                                                                                                                                          |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normative knowledge base    | Each project's binding criteria are buried in its own technical file and lab mix designs. Extracting and structuring these correctly requires domain access. Every project we process makes the library deeper. |
| State system integration    | Feeding INFOBRAS in the Comptroller's required format is a compliance feature, not a convenience feature. Foreign platforms will not build it.                                                                  |
| Channel through credibility | Sector trust in Peru is personal. A civil engineer with institutional reputation opens doors that marketing spend cannot.                                                                                       |

### 5.4 The wedge and the ladder

| Stage  | What we sell                                       | To whom                    | Why they buy                                    |
|--------|----------------------------------------------------|----------------------------|-------------------------------------------------|
| Wedge  | Quality record capture and dossier generation      | Quality Specialist         | Personal legal protection, stops chasing people |
| Step 2 | Payment certificate backing package                | Site Resident / contractor | Unblocks monthly cash flow                      |
| Step 3 | Real-time compliance visibility and INFOBRAS feed  | State entity / supervision | Audit readiness, sanction avoidance             |
| Step 4 | Multi-project quality intelligence across a region | Regional Government        | Portfolio-level control of stalled-project risk |

## 6. WHAT WE ARE BUILDING

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*Protokol captures quality data where the work happens, validates it against the legally binding standard, and produces the document package the state requires — automatically.*

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### 6.1 Three layers

#### Layer 1 — Capture (field)

A progressive web app. No install from an app store, no account creation, opens from a link, works on a low-end Android phone with intermittent signal. The field technician records what they just did in under five minutes.

- Guided forms per activity type, not free-form data entry
- Camera capture with GPS coordinates and tamper-resistant timestamp
- Offline-first: queues locally, syncs when signal returns
- Real-time validation feedback — the technician knows immediately if a value is out of range

#### Layer 2 — Intelligence (backend)

Maintains the true state of the project at all times and enforces the rules.

- Validates every submitted value against project-specific criteria loaded as data
- Automatically opens a non-conformance record when a value fails
- Tracks the release state of every panel across every protocol stage
- Manages the deferred workflow: cylinder strength results arriving 7 and 28 days after the pour
- Immutable evidence store — records are append-only with integrity hashing

#### Layer 3 — Output (office and state)

- Generated protocol PDF per record, in the format supervision accepts
- Live status view for the quality manager — what is released, pending, or non-conforming
- Quality dossier compilation for contract closure
- Export in INFOBRAS-compatible structure

### 6.2 What WhatsApp is and is not

WhatsApp is the notification and escalation channel — not the data capture channel. Notifications go out when a protocol completes, when a non-conformance opens, and when a record is overdue. Data entry happens in the PWA, because photo quality, GPS metadata and structured fields matter legally and WhatsApp degrades all three.

## 7. PHASE 0 — THE PILOT BUILD

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► *Scope discipline: Phase 0 exists to answer one question, not to be a product.*

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*Does the field team record quality data through Protokol without the quality manager chasing them?*

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## 7.1 In scope

- PWA with guided forms for three activities: concrete, survey, compaction
- Backend API with the four endpoints in Section 8
- Immutable record storage with photo handling
- Project-specific validation criteria loaded as data
- Protocol PDF generation
- WhatsApp notifications to the quality manager
- Deferred cylinder result entry (7 and 28 day strength)
- Simple status view — a synced spreadsheet is acceptable for Phase 0

## 7.2 Explicitly out of scope

- Full web dashboard — deferred to Phase 1
- Native mobile app — the PWA is sufficient
- Multi-project support — one project, hardcoded, is fine
- INFOBRAS integration — design for it, do not build it yet
- Billing, subscriptions, self-service onboarding
- Complex authentication — see Section 9 on identity

# 8. API CONTRACTS

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► *These contracts define what crosses the boundary. Everything inside is your call — stack, framework, patterns, deployment.*

## 8.1 POST /protocols

Submitted when the technician completes a form. Server validates against stored criteria and returns the verdict.

```
// Request
{
  "project_id": "AY-728-001",
  "device_token": "dvc_8f3a...", // issued once per device
  "technician_pin": "1234",
  "activity": "CONCRETE", // CONCRETE | SURVEY | COMPACTION
  "recorded_at": "2026-09-11T08:47:00-05:00",
  "gps": { "lat": -13.1588, "lng": -74.2236 },
  "panel": "15",
  "chainage": "0+144",
  "measurements": {
    "slump_cm": 8,
    "mixer_id": "6D37",
    "delivery_note": "GR-00412", // legal delivery document
    "cylinders_cast": 2,
    "design_fc": 210
  },
  "photo_ids": ["ph_a91c...", "ph_b22e..."],
  "notes": ""
}
```

```
// Response
{
  "protocol_id": "PRT-20260911-0847-015",
  "verdict": "PROVISIONAL_PASS", // PASS | PROVISIONAL_PASS | FAIL
  "checks": [
    { "field": "slump_cm", "expected": "7.6 - 10.2", "actual": 8, "result": "PASS" }
  ],
  "nonconformance_id": null,
  "pdf_url": "https://.../PRT-20260911-0847-015.pdf",
}
```



```

    "pending":      ["CYLINDER_7D", "CYLINDER_28D"]
  }

```

► *PROVISIONAL\_PASS matters: a concrete pour cannot be finally certified until cylinder strength results arrive weeks later. Modeling this as a two-stage verdict is a domain requirement, not a nicety.*

## 8.2 POST /protocols/{id}/cylinder-result

Submitted when the lab returns 7-day or 28-day compressive strength. Converts a provisional pass into a final verdict, or opens a non-conformance.

```

{
  "age_days":      28,
  "cylinder_code": "P-2026-0847-A",
  "strength_kgcm2": 231,
  "lab":           "AKHISE",
  "report_photo_id": "ph_c77d..."
}

```

## 8.3 GET /projects/{id}/status

Returns the current release state of the whole project. Drives the status view and, later, the dashboard.

```

{
  "project_id":  "AY-728-001",
  "as_of":       "2026-09-11T09:15:00-05:00",
  "summary": {
    "total_expected": 84,
    "passed":         71,
    "provisional":    6,
    "failed":         3,
    "missing":        4
  },
  "protocols": [ /* ... */ ]
}

```

## 8.4 GET /projects/{id}/dossier

Compiles every protocol, photo, lab result and non-conformance into the quality dossier package.

```

{
  "dossier_url":  "https://.../AY-728-dossier.pdf",
  "generated_at": "2026-09-11T09:30:00-05:00",
  "protocols_included": 84,
  "open_nonconformances": 1,
  "completeness_pct": 94.2
}

```

# 9. DATA MODEL AND IRREVERSIBLE DECISIONS

## 9.1 Core tables

| Table           | Purpose                                                | Critical fields                                                                                                                         |
|-----------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| projects        | One row per construction project                       | id, name, contract_number, entity, execution_mode                                                                                       |
| protocols       | The central artifact. One per recorded activity.       | id, project_id, activity, recorded_at (with timezone), gps, panel, chainage, measurements (JSON), verdict, device_token, integrity_hash |
| criteria        | Project-specific validation thresholds, loaded as data | project_id, activity, field, operator, min, max, source_reference                                                                       |
| cylinders       | Concrete test specimens and their deferred results     | protocol_id, cylinder_code, cast_date, test_date, age_days, strength, lab                                                               |
| nonconformances | Failures requiring corrective action                   | protocol_id, description, status, corrective_action, closed_at                                                                          |
| photos          | Evidence images with metadata                          | id, protocol_id, storage_key, gps, captured_at, hash                                                                                    |

|               |                      |                                                              |
|---------------|----------------------|--------------------------------------------------------------|
| technicians   | Field users          | id, project_id, name, pin_hash, device_token, whatsapp, role |
| notifications | Outbound message log | recipient, event_type, payload, sent_at, status              |

## 9.2 Four decisions that cannot be undone later

Everything else in this build is reversible in days or weeks. These four are not. Please confirm your position on each before writing code.

| Decision                                    | Why it is irreversible                                                                                             | Our recommendation                                                                                                 |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Validation criteria in code vs. in database | If hardcoded, the entire product gets designed around single-project assumptions. Retrofitting touches everything. | Store as data from day one. Each project loads its own criteria. Non-negotiable for the business model.            |
| Timestamp handling                          | Records stored without explicit timezone become legally ambiguous forever. Cannot be repaired retroactively.       | Always store UTC with explicit offset. Display in project-local time.                                              |
| Technician identity mechanism               | The legal value of every record depends on proving who submitted it. Past records cannot be re-signed.             | PIN for UX, plus a persistent device token issued once. Store device + PIN + GPS + integrity hash on every record. |
| Photo storage references                    | Thousands of photos with provider URLs embedded in immutable records make migration painful.                       | Store our own opaque identifier. Resolve to a URL at read time. Never persist a vendor URL inside a record.        |

## 9.3 Validation criteria — real values from the pilot project

These come from the project's binding technical file and the accredited lab's mix design. They load as data per project, never as constants.

| Activity   | Field               | Pass criterion                                                                          | Source                           |
|------------|---------------------|-----------------------------------------------------------------------------------------|----------------------------------|
| CONCRETE   | slump_cm            | Per the mix design marked as current for this project — plastic consistency range       | Accredited lab mix design report |
| CONCRETE   | design_fc           | Per the activity's specified strength (140 / 175 / 210 / 245 / 280 kg/cm <sup>2</sup> ) | Technical file specifications    |
| CONCRETE   | cylinders_cast      | Minimum 2 per pour above 4 m <sup>3</sup>                                               | EG-2013                          |
| CYLINDER   | strength_kgcm2      | ≥ design f'c at 28 days                                                                 | EG-2013                          |
| COMPACTION | compaction_pct      | ≥ 100% of Modified Proctor                                                              | Project quality plan             |
| COMPACTION | moisture_deviation  | Within ±1.5% of optimum                                                                 | Project quality plan             |
| COMPACTION | sub_base_thickness  | ≥ 20 cm                                                                                 | Technical file                   |
| COMPACTION | base_thickness      | ≥ 25 cm                                                                                 | Technical file                   |
| SURVEY     | elevation_deviation | ≤ 1 cm from design                                                                      | Project quality plan             |

► *Kenny will deliver the exact numeric thresholds as a table before build starts. Do not infer them from general standards — they are project-specific and getting them wrong produces false non-conformances.*

## 10. ACCEPTANCE CRITERIA

Written as testable statements. Each one is verifiable without interpretation.

| #  | Requirement                                                                                                                                                                         |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1 | WHEN a technician submits a protocol with all required fields, THE SYSTEM SHALL persist an immutable record and return a verdict within 5 seconds under normal connectivity.        |
| R2 | WHEN a submitted measurement falls outside the configured criterion, THE SYSTEM SHALL create a linked non-conformance record and notify the configured recipient within 60 seconds. |
| R3 | WHEN the device has no connectivity, THE SYSTEM SHALL queue the submission locally and sync automatically once connectivity returns, without user action.                           |
| R4 | WHEN a concrete protocol is submitted, THE SYSTEM SHALL mark it PROVISIONAL_PASS and remain pending until cylinder results at 28 days are recorded.                                 |
| R5 | WHEN cylinder strength below design f'c is recorded, THE SYSTEM SHALL change the protocol verdict to FAIL and open a non-conformance.                                               |

|     |                                                                                                                                                                     |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R6  | THE SYSTEM SHALL store every timestamp in UTC with explicit offset and never accept a client-supplied timestamp without recording server receipt time alongside it. |
| R7  | THE SYSTEM SHALL reject any attempt to modify a persisted protocol record. Corrections occur through a new linked record, never by overwriting.                     |
| R8  | WHEN a protocol is completed, THE SYSTEM SHALL generate a PDF containing all recorded values, photos, technician identity, timestamp and chainage.                  |
| R9  | WHEN a photo is captured, THE SYSTEM SHALL store its GPS coordinates and capture time as metadata independent of the image file.                                    |
| R10 | WHEN an expected protocol has not been submitted 4 hours after its scheduled activity, THE SYSTEM SHALL notify the quality manager.                                 |

## 11. HOW WE WORK TOGETHER

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### Authority

- Stack, architecture, libraries, patterns, infrastructure: your call. Kenny does not have an opinion and will not develop one.
- What gets built, in what order, and what is in scope: Kenny's call. You can object on technical cost and that objection gets recorded.
- If we deadlock: the reversible option wins. If neither is reversible, we defer until we have more information.

### Cadence

- Weekly deploy to a link Kenny can open on a phone. Whatever state it is in.
- Kenny tests within 48 hours and responds in writing: what works, what does not, what is missing.
- One 30-minute call per week. Fixed agenda: blockers, then decisions, then demo.

### Domain questions

- Never guess about construction domain. If a decision depends on how the physical or legal process works, flag it as a DOMAIN BLOCKER and move to another task.
- Kenny responds to domain blockers within 24 hours, or within 48 if he needs to consult the field.
- Every domain answer gets written into this document, not just the chat. This document is the source of truth.

### Decision records

Every architectural decision gets a short file in the repository: context, decision, consequences, and reversibility rating. Keeps both of us honest about what was decided and why.

### Definition of done

Deployed and reachable by link · meets its acceptance criteria · Kenny tested it on a real phone · the architectural decision is recorded.

## 12. WHAT I NEED FROM YOU

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Read the document, then answer these. Nothing gets built until we have these settled.

| # | Question | Why it matters |
|---|----------|----------------|
|---|----------|----------------|

|   |                                                                                                            |                                                                                      |
|---|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1 | How many hours per week can you realistically give this alongside your other work?                         | Determines whether Phase 0 is 4 weeks or 10. I would rather plan honestly than miss. |
| 2 | Which parts of this document did not make sense?                                                           | The domain is foreign and I will have explained something badly. Tell me where.      |
| 3 | Do you agree with the four irreversible decisions in Section 9.2, or would you do any of them differently? | These are the only ones we cannot fix later.                                         |
| 4 | Anything in the acceptance criteria that is technically unreasonable or ambiguous?                         | Better to fix now than to build the wrong thing.                                     |
| 5 | What infrastructure do you want, and what does it cost per month?                                          | I need to budget it.                                                                 |
| 6 | Is there something you would build first that I did not propose?                                           | You may see a better sequencing than I do.                                           |